# TaskVine / VineReduce / vine-cms-analysis-stack

## Module 0 - Intro

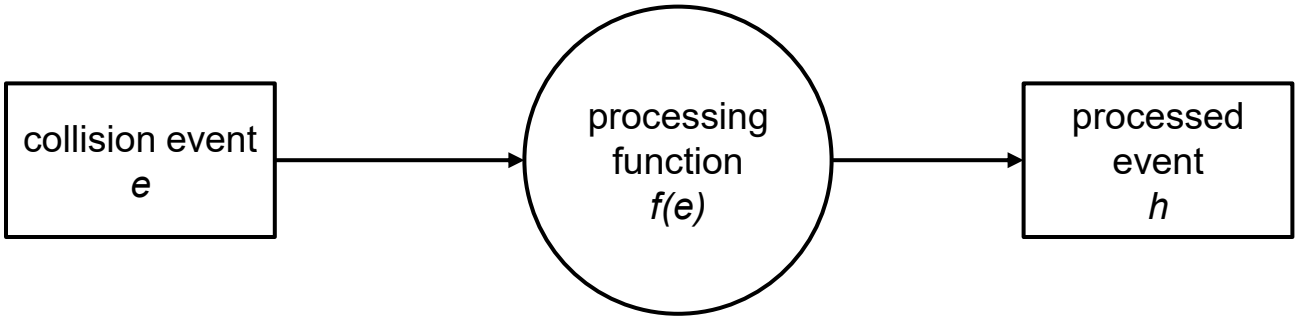
3-hour hands-on tutorial for HEP graduate students

# Scaling from First Principles

Apply processing function **f** to event **e** to get result **h**.

f: E → H

h = f(e)

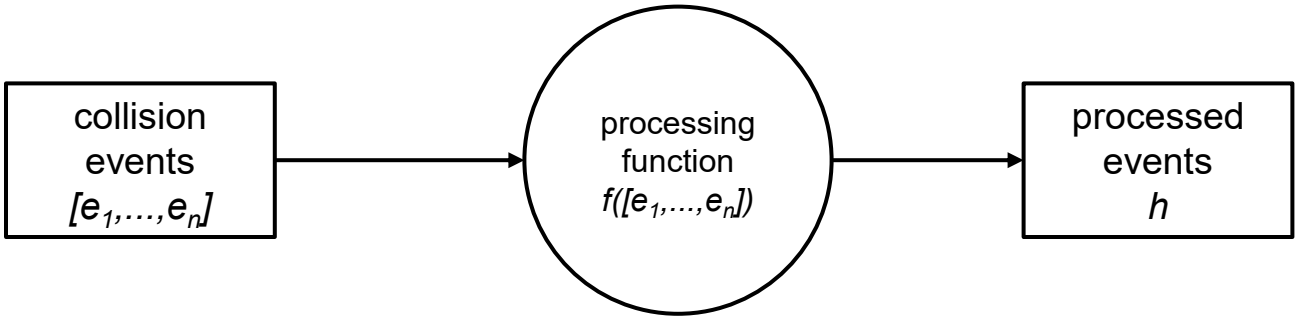


# Scaling from First Principles

Almost always we process more than one event per function call.

$f: E \rightarrow H, \quad +: H \times H \rightarrow H$

**$f([e_1,...,e_n]) = f(e_1) + ... + f(e_n)$**



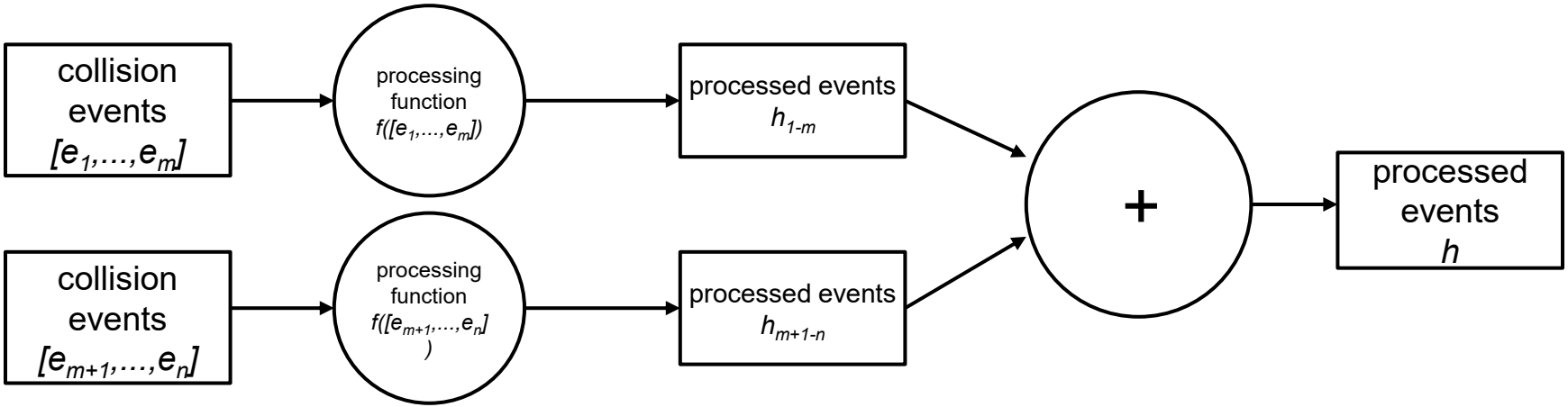

If $f([e_1,...,e_n])$ can run efficiently in a single machine, we are done.

# Scaling from First Principles

The + only comes into play when we want to scale the computation, and only if it is associative.

(It is like map-reduce, but the reduce part is only important when it allows us to scale.)

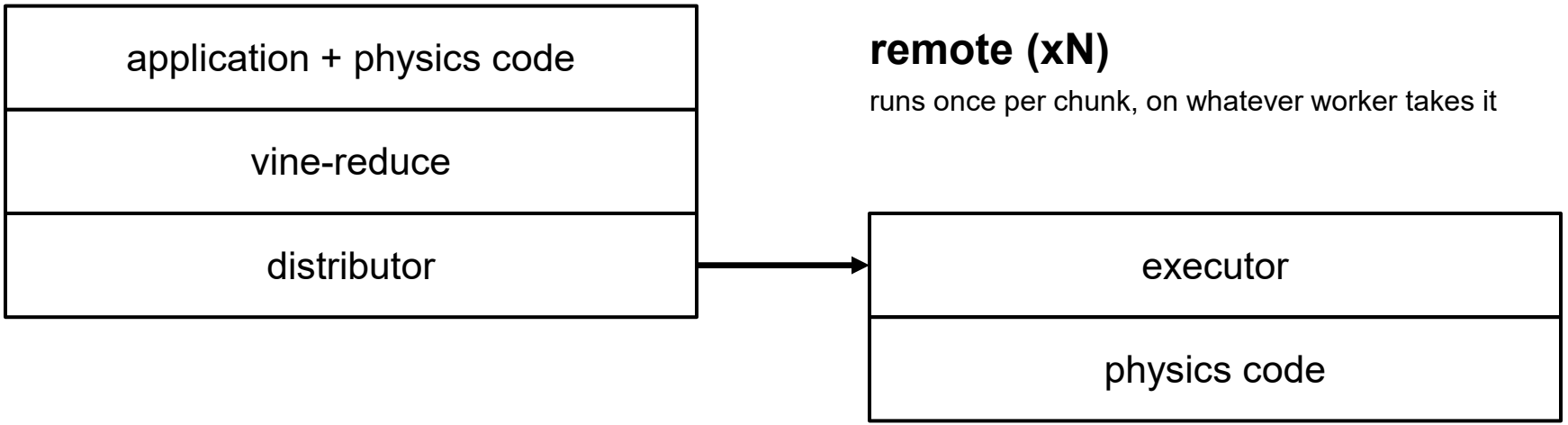
**$f([e_1,...,e_n]) = f([e_1,...,e_m]) + f([e_{m+1},...,e_n]) = f(e_1) + ... + f(e_n)$**



# The shape of the stack

## local (x1)

runs once, on the machine you are logged into



# What we are running

- ADL benchmark Q6: a top-quark trijet selection on CMS Open Data
- Small, but the same shape as a real
- Written in coffea or RDataFrame; you pick one




Partial results are added together until one result per dataset remains.

# Why VineReduce, alongside coffea/RDataFrame

## Parallelism

Owned by VineReduce

- Split a dataset into chunks
- Map the processor over chunks
- Reduce outputs back together

## Distribution

Handed to a distributor (today: TaskVine)

- Schedule chunks onto machines
- Transfer code and data

## Execution

Owned by coffea/RDataFrame

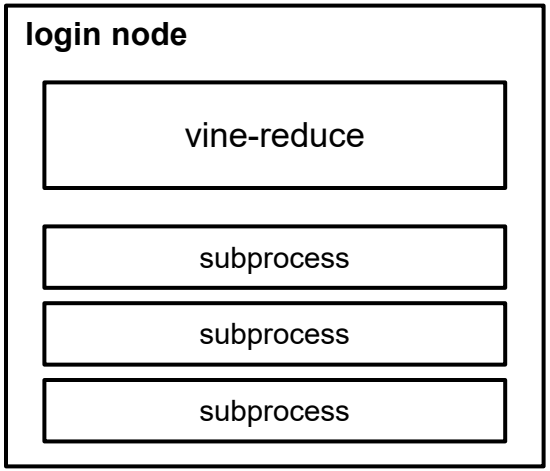
- Execute code on remote workers

**The analysis code never changes. Only what it is handed to does.**

# Three places for execution

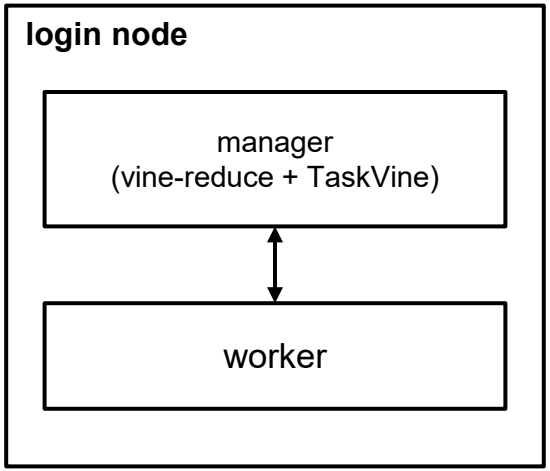
## 1 Local distributor

Module 2



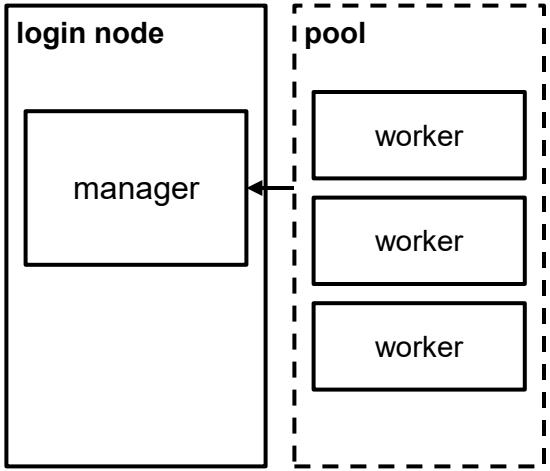

## 2 Local TaskVine

Module 3



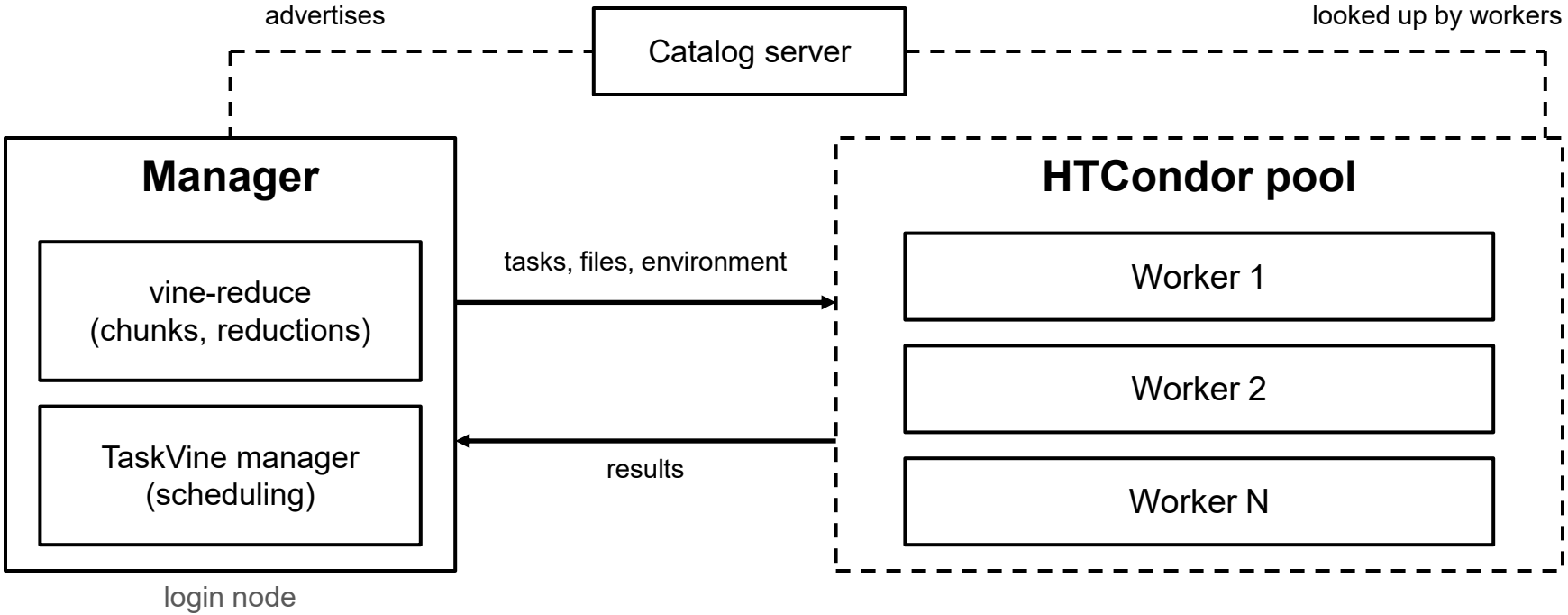

## 3 HTCondor pool

Module 4




**The analysis code is the same in all three. Only what it is handed to changes.**

# Manager and workers




Workers find the manager through the catalog, connect to it, and read input data from storage (xrootd or a shared filesystem).

# What the manager lets you control

| | | |
|---|---|---|
| Module 5 | **Resources and chunk size** | What a task asks for, and how big it is |
| Module 6 | **Checkpoints and restart** | Stop a run, resume without redoing work |
| Module 7 | **result_postprocess** | Big outputs bypass the manager |
| Module 8 | **Multiple processors** | Several analyses in one run |
| Module 9 | **failure_proportion** | Keep going when some chunks fail |

# Where to get help

## NDCMS Mattermost

`https://mattermost.web.cern.ch/ndcms/channels/cctools`

## Cooperative Computing Lab

`https://cclnd.cse.nd.edu`

## TaskVine documentation

`https://cctools.readthedocs.io/en/stable/taskvine`

## GitHub

`github.com/cooperative-computing-lab: vine-reduce, vine-cms-analysis-stack, cctools`